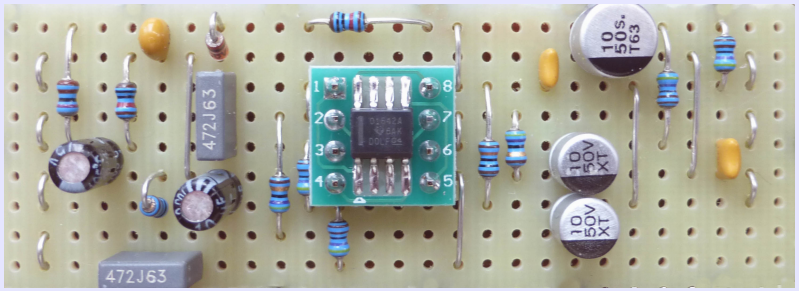
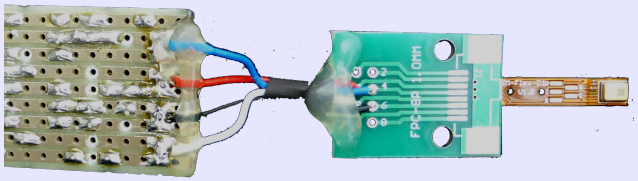
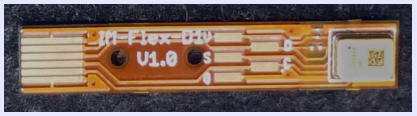
MEMS: 48v Phantom Powered Differential Ultrasonic Mic Interface

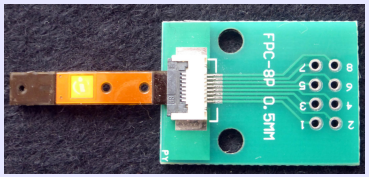
- Interfaces a MEMS capsule to a 48v phantom powered mic pre-amp
- Capsule is INFINEON type IM73A135 (differential analogue output)
- Designed to amplify ultrasonic bat echolocation calls
- Simple stripboard layout

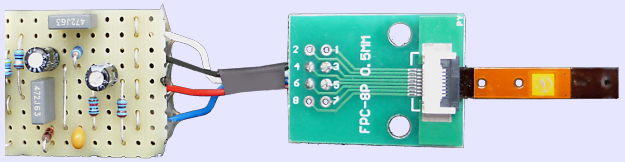








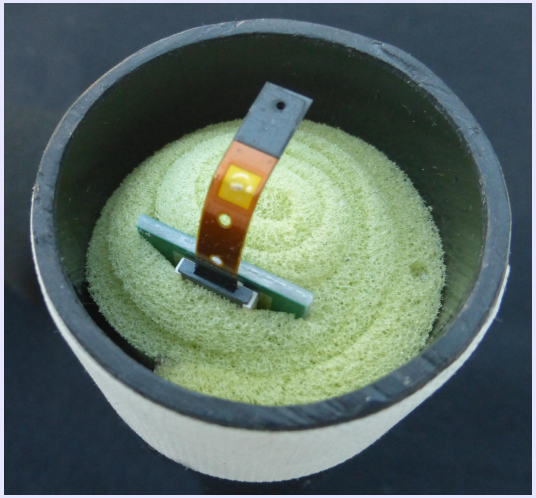


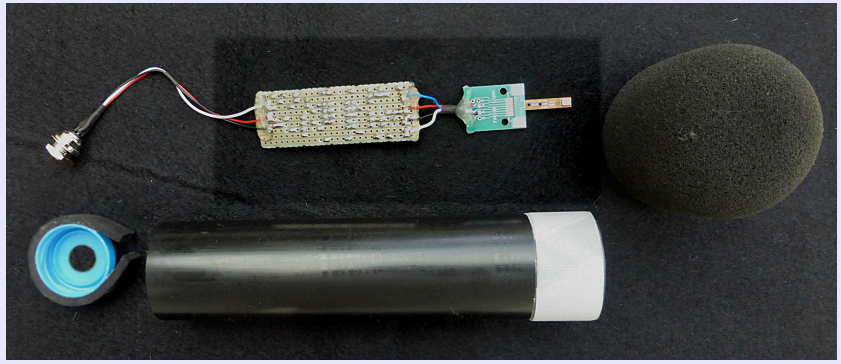


- Mic Capsule: https://www.infineon.com/assets/row/public/documents/24/45/infineon-microphoneflexevaluationkit-productbrief-en.pdf

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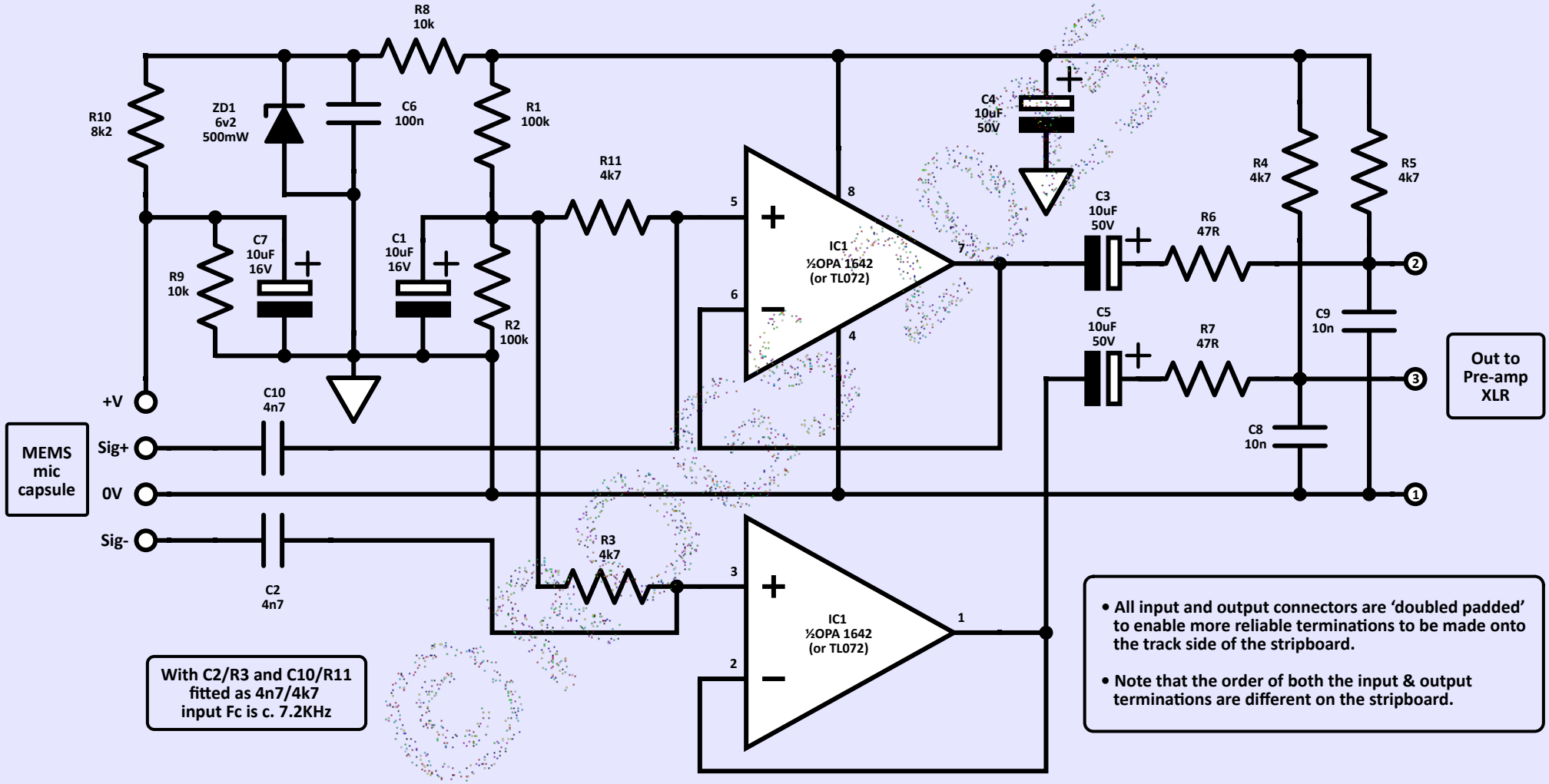




- Mic Housing - 180mm x 40mm dia. Polypropylene pipe
- Output connector - Mini XLR type - https://www.rean-connectors.com/en/product/rt3mpr-1

https://zoom-ultrasonic.github.io/

• MEMS 48v Phantom Powered Ultrasonic Mic Pre-amp Mk.4 - SCHEMATIC •



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MEMS 48v Ultrasonic Microphone Pre-amp: Mk4
(Differential Input and Balanced Output)

Overview:

The circuitry shown in the schematic is designed to interface an **Infineon IM73A135** MEMS microphone capsule with a conventional balanced microphone pre-amplifier, equipped with 48v phantom power and a 3 pin XLR connection.

The circuit is specifically designed to record ultrasonic 'echolocation' signals emitted by bats, and as such does not have a conventional audio range frequency response.

The strip-board layout that the components are fitted to is intended to be mounted into a lightweight enclosure – typically a short 180mm length of 40mm diameter polypropylene pipe. This will allow for the creation of a lightweight microphone enclosure, suitable for mounting onto a boom pole.

Circuit description:

The active device used for this project can be either the Texas Instruments OPA1642 operational amplifier (op-amp) for best performance, or a TL072 for simpler construction.

Both allow for a low distortion impedance converting amplifier to be constructed.

(The OPA1642 has better noise figures, but does require an SOIC to DIL adaptor).

The op-amps are configured in this project as non-inverting, unity gain buffer amplifiers. This is achieved by connecting the amplifier outputs directly to the associated inverting input.

The op-amps are powered from the 48v phantom power of the attached microphone pre-amp. Resistors R4 and R5 are connected to pins 3 and 2 of the XLR input connector. The other ends of both resistors are joined and - together with the decoupling capacitor C4 - allow a DC voltage of c.20v to be applied to pin 8 of the dual op-amp. The actual DC value is not critical, and the values of R4 and R5 are selected to ensure the maximum permitted supply to the op-amp cannot be exceeded.

The op-amps requires a 'dual' voltage supply, and this is provided by the creation of a 'half rail' voltage of around 10v by the voltage divider R1 and R2, together with the decoupling capacitor C1.

This 'half rail' voltage is required to bias the op-amps, so that the output can swing symmetrically around this reference voltage. It is fed to the non-inverting inputs of the op-amps via R3 and R11.

The ' **Sig+ '** input from the microphone capsule itself is fed to the non-inverting input of one op-amp via C10. The values of R11 and C10 are selected to provide a high pass filtered input signal, where the 'cut off' (-3dB) frequency is set at c.7KHz. This will permit the lower frequencies of bat echolocation signals to be recorded, but will attenuate the vast majority of 'normal' audio frequency signals, which only serve to confuse with this type of recording.
The ' **Sig- '** from the capsule is fed similarly to the second op-amp, via C2.

The MEMS microphone capsule requires a separate DC supply to be applied to the '**+V** ' terminal on the schematic. This DC should have a value within the range 1.5 and 3v, and in this case the value is selected to be c. 2.7v.

A 6v2 Zener diode is fed with DC from the R4/R5 summing junction via R8. This point is de-coupled by C6 to remove any Zener noise.

A second voltage divider - formed by R10 and R9, and decoupled by C7 – provides an appropriate 2.7v DC supply to the MEMs capsule.
The Zener is included to both regulate and limit the maximum DC supply supplied to the capsule, which can be sensitive to over voltage.

The **Sig+** op-amp ultrasonic audio output is taken from pin 7 and fed, via C3 and R6, to pin 2 (hot) of the XLR connector. (C3 is included to isolate the op-amp DC level from the phantom power present on the XLR connector).

The **Sig-** op-amp ultrasonic audio output is taken from pin 1 and fed, via C5 and R7, to pin 3 (cold) of the XLR connector. (C5 is included to isolate the op-amp DC level from the phantom power present on the XLR connector).

C8 an C9 are fitted to help supress any stray RF interference that may be present in the vicinity of microphone operation.

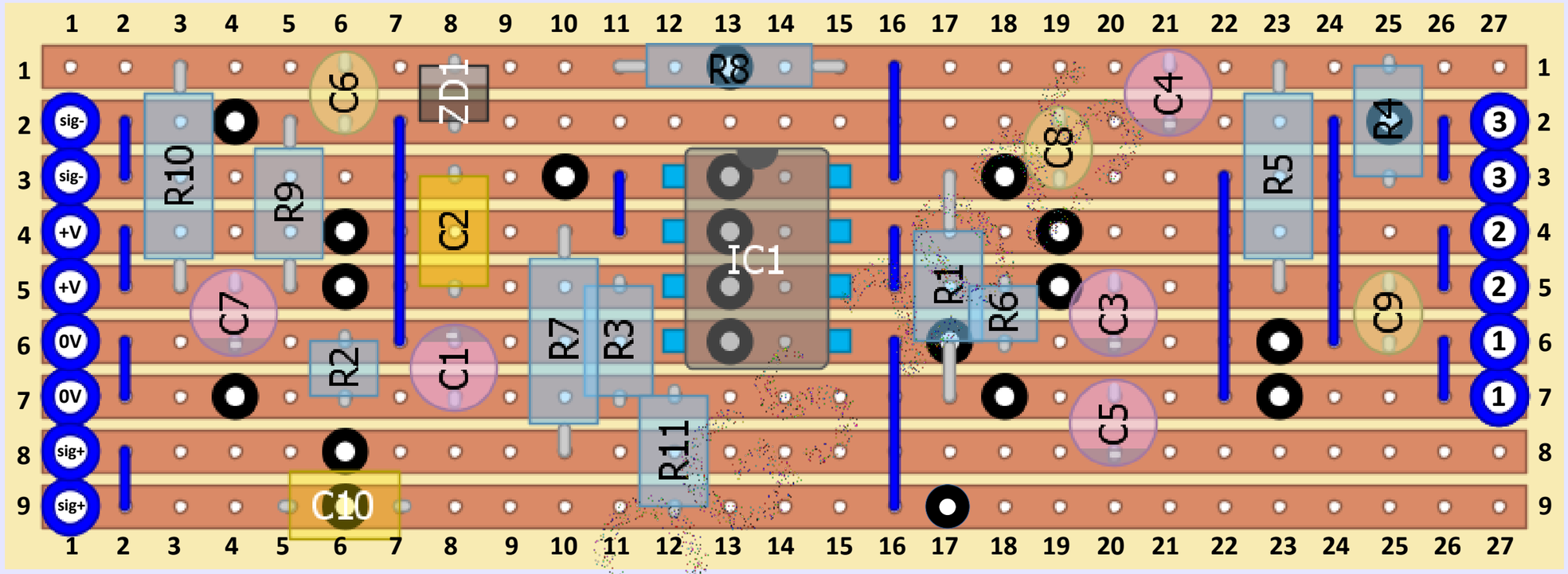
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MEMS 48v Phantom Powered Ultrasonic Mic Pre-amp Mk4 - Stripboard layout

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From MEMs Capsule



To Mic pre-amp (Via Mini XLR)

Stripboard

- FC2 or FC4 (KEMO E005)
- 9 (copper tracked) rows
- 27 columns
- 21 x 'spin off' track cuts at black and white dot locations
- 14 x TCW links at blue line locations (25 gauge or similar)
- *The 'double' termination reference numbers allow for external wire connections to be made between tracks – on the track side - for a more robust and reliable connection.*

Parts List

- Capacitors:

- *'elec' = electrolytic*
- *'mlc' = multi layer ceramic*
- *'PET' = Polyethylene film*

- C1, 7 - 10uF 16v elec
- C2, C10 - 4.7nF 63v PET
- C3, 4, 5 - 10uF 50v elec
- C6 - 100nF 63v mlc
- C8, C9 - 10nF 63v mlc

- *All resistors are MF12 1/8th watt metal film:*

- R1, 2 - 100k
- R3, 4, 5, 11 - 4k7
- R8, 9 - 10k
- R10 - 8k2
- R6, 7 - 47R

- IC1 OPA1642 (Pin1 is column 12, row 3)

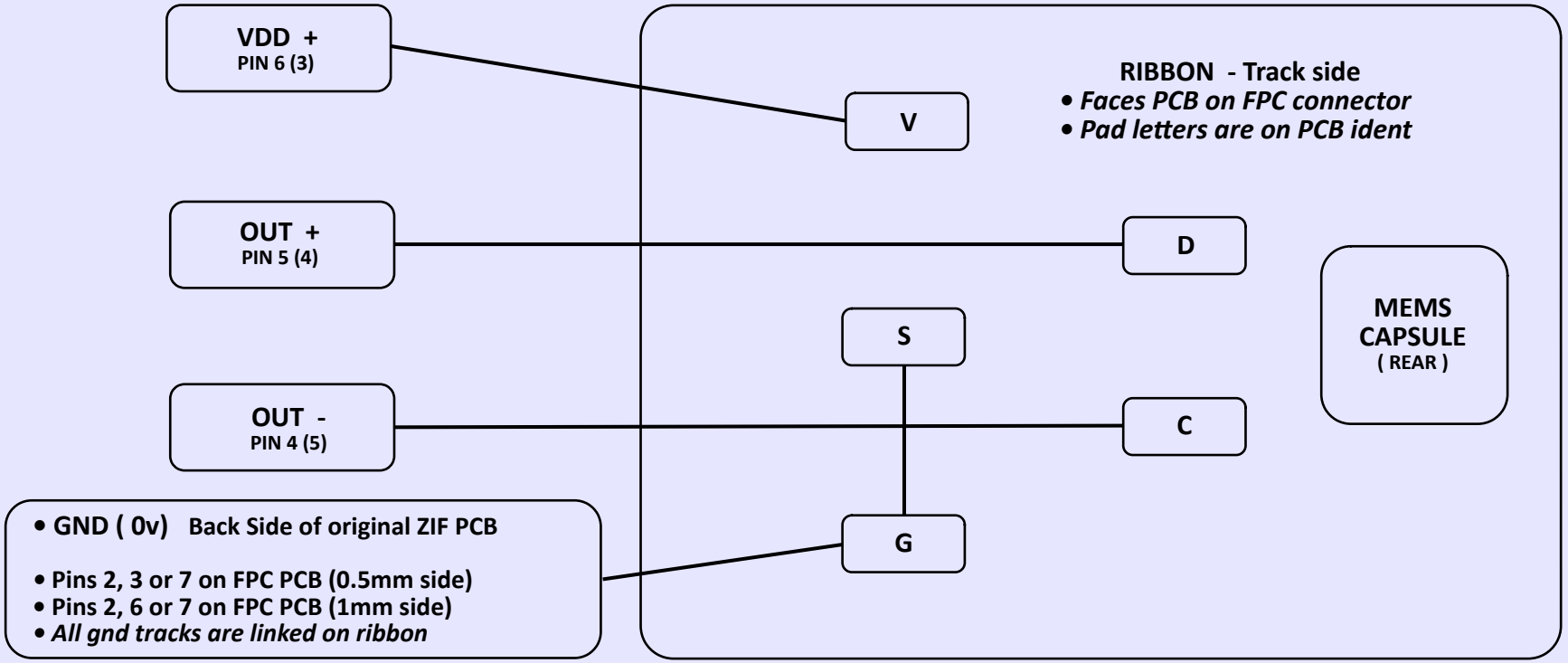
- Use SOIC DIP adaptor for IC1
  (TL072 can be used as a DIP equivalent op-amp, but will have a higher noise level)

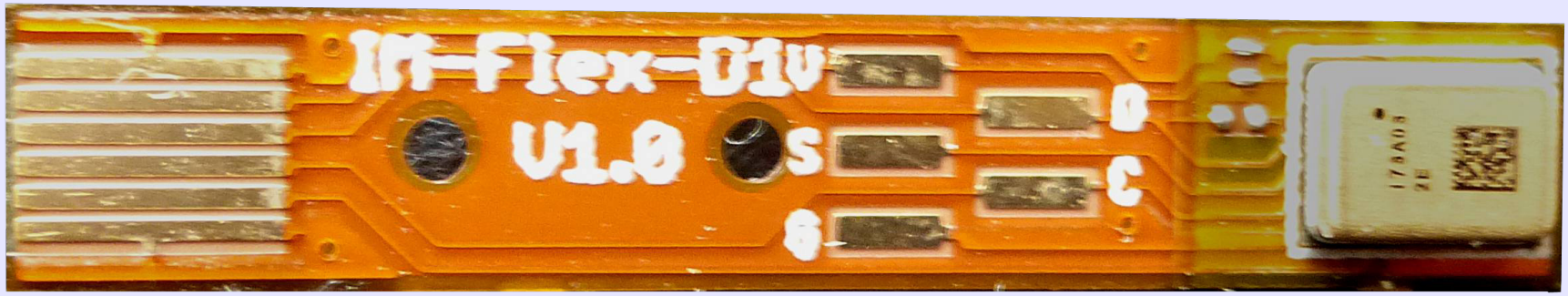
- ZD1 6v2 Zener diode 500mW

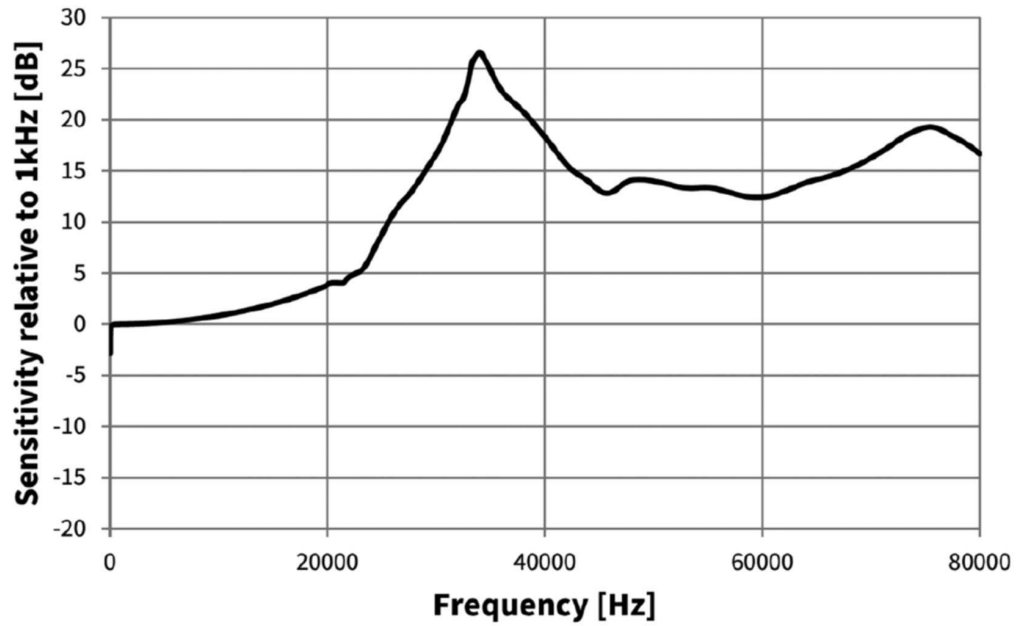
INFINEON IM73A135 – CONNECTIONS TO RIBBON TERMINATION PADS

- Solder pad numbers on original ZIF PCB
- Also hole numbers on FPC 8P 0.5mm adaptor - when viewed from 0.5mm side
- Numbers in brackets ( ) are numbers on FPC 8P 0.5mm adaptor viewed from 1mm side

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Infineon IM73A135
Ultrasonic Free Field Response

The graph above shows the Ultrasonic response of the Infineon IM73A135 MEMs microphone capsule…..

As you can see, it's not very linear! .. An erratic response like this is difficult to correct in the analogue mode - in real time - without complex filters being applied.

However, processing pre-recorded files in the digital mode, by the sequential application of the 2 equalisation processes described on the right, can make frequency response correction a reasonably simple procedure.

The detailed notes on the right describe a script used in Adobe Audition to perform the task on the 32 bit float files recorded with a Zoom F3 recorder. (Other DAW/recorder combinations should offer similar functions).

Adobe Audition Script for INF135 MEMs mic HF frequency response correction

- '**BATX LF cut off' Graphic EQ preset**: *30 band settings*
  Range: 120dB.... ***All set to -60dB except****:*

- 10KHz •12.5KHz •16KHz •20Khz •25KHz bands -
  - which are set at 0dB.

- **'INF135' Parametric EQ preset filter**:

- *Range setting: 10kHz to 80KHz* • *Constant Q* • *Gain: 0dB*

- *3 Band settings (upper 2 bands not used)*

- 12KHz: Q=2 +6dB • 28KHz: Q=10 -3dB • 33KHz: Q=10 -15dB

Script: *The following actions are applied in sequence:*

- Apply **BATX LF cut off**
- Apply **INF135** Parametric EQ filter
- Convert 32 bit float file to 16 bit
- End of Script

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INFINEON IM73A135 --- MEMS capsule - maximum output voltage

- Sensitivity: -38dBV
- Convert to volts (ref: 1Pa/94dB SPL): V 94dB = 10^-38/20 = 12.6mV RMS **(35.6mV p-p)**
- Max SPL=135dB = 41dB above 94dB ref.
- Pressure multiplier = 10^ (41/20) = 10^2.05 = 112Pa
- Output voltage @ 135dB SPL = 12.6mV x 112 = 1.41V RMS (+5.2 dBu) * see notes below
- Vp-p = 1.41 x 1.414 x 2 = **4Vp-p**

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* Zoom F3 pre-amp characteristics

- max. mic input voltage for Zoom F3 = +4dBu (1.23V RMS = **3.4V p-p**) (lower than max. mic output level)
- max. line input voltage for Zoom F3 = +24dBu (12.28V RMS = **34.72V p-p**)
- Use of line input will guarantee that the max. mic output level will determine the clipping point, not the Zoom F3 pre-amp.

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